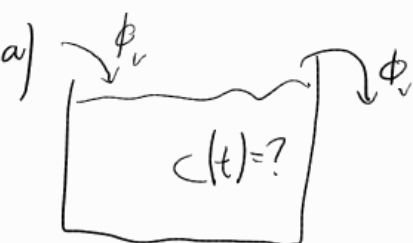


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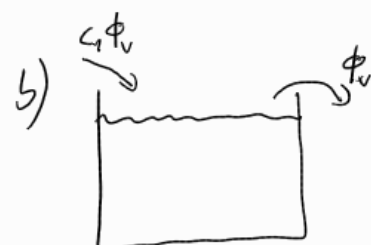
$$c = \frac{m}{V}$$

$$dm = -c \phi_v dt \quad / : V$$

$$\frac{dc}{c} = - \frac{\phi_v}{V} dt$$

$$\ln \frac{c}{c_0} = - \frac{\phi_v}{V} t$$

$$c(t) = c_0 e^{-\frac{\phi_v}{V} t}$$



$$dm = -c \phi_v dt + c_1 \phi_v dt$$

$$dc = (c_1 - c) \frac{\phi_v}{V} dt$$

$$\dot{c} = (c_1 - c) \frac{\phi_v}{V}$$

$$c_H = c_0 e^{-\frac{\phi_v}{V} t}$$

nastavek za c_p : $c_p = A e^{-\frac{\phi_v}{V} t} + B$

$$\dot{c}_p = -\frac{\phi_v}{V} A e^{-\frac{\phi_v}{V} t} = (c_1 - A e^{-\frac{\phi_v}{V} t} - B) \frac{\phi_v}{V}$$

$$B = c_1$$

$$\rightarrow c(t) = c_H + c_p = c_0 e^{-\frac{t}{\tau}} + A e^{-\frac{t}{\tau}} + c_1$$

$$c(0) = c_0 \rightarrow c_0 = c_0 + A + c_1 \rightarrow A = -c_1$$

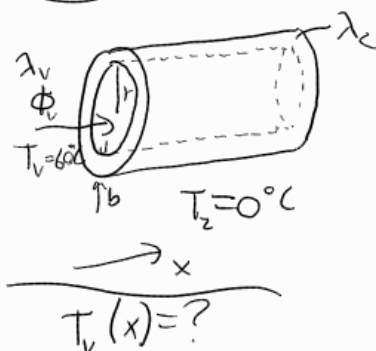
$$\Rightarrow \boxed{c(t) = c_0 e^{-\frac{t}{\tau}} - c_1 e^{-\frac{t}{\tau}} + c_1}$$



$$h(t) = ?$$

Pride neki zlo grdega

7.11 Kodre 123/5



(voda pošljemo skozi cev, T_v je konst. pri dol. x)

$$\rho c S_0 dx \frac{dT}{dt} = - \frac{\lambda}{b} (T - T_s) dS - S_0 (j(x+dx) - j(x))$$

$$\rho c \frac{dT}{dt} dx = - \frac{2\lambda}{br} (T - T_s) dx - j(x+dx) + j(x)$$

$$-\frac{dj}{dx} = \rho c \frac{dT}{dt} + \frac{2\lambda}{br} (T - T_s)$$

$$-\frac{dj}{dx} = \rho c \frac{d\tilde{T}}{dt} + \frac{2\lambda}{br} \tilde{T} =$$

$$= \rho c \left(\frac{\partial \tilde{T}}{\partial t} + (\vec{v} \cdot \nabla) \tilde{T} \right) + \frac{2\lambda}{br} \tilde{T}$$

(izčemo stac. st.)

$$\frac{dT}{dt} = \frac{\partial T}{\partial t} + (\vec{v} \cdot \nabla) T$$

$$j = -\lambda_v \nabla T$$

$$dj = -\lambda_v \frac{dT}{dx}$$

$$+\lambda_v \frac{d^2 \tilde{T}}{dx^2} = \rho c v \frac{d\tilde{T}}{dx} + \frac{2\lambda}{br} \tilde{T}$$

$$\lambda_v \tilde{T}'' - \rho c v \tilde{T}' - \frac{2\lambda}{br} \tilde{T} = 0$$

$$D \tilde{T}'' - T' - \frac{2\lambda}{rb\rho c v} \tilde{T} = 0$$

$\frac{1}{x_0}$

1) $D=0$ (topl. se prevaja le v shreg)

$$\tilde{T} = \tilde{T}_0 e^{-\frac{x}{x_0}}$$

2) $\lambda=0$

$$D \tilde{T}'' - \tilde{T}' = 0$$

$$\text{naključek: } \tilde{T} = A e^{kx}$$

$$D k^2 A e^{kx} - k A e^{kx} = 0 \rightarrow k=0$$

$\tilde{T} = \text{konst.}$

$$D k - 1 = 0$$

$$k = \frac{1}{D} \rightarrow \tilde{T} = A e^{\frac{x}{D}}$$

Ni smiselno, ker $\tilde{T}$ narašča

(bi bilo smiselno, če T na desni višja)

3) cela rešitev:

$$Dk^2 Ae^{kx} - kAe^{kx} - \frac{A}{x_0} e^{kx} = 0$$

$$Dk^2 - k - \frac{1}{x_0} = 0$$

$$k_{1,2} = \frac{1 \pm \sqrt{1 + 4 \frac{D}{x_0}}}{2D}$$

robni pogoja: $\tilde{T}(x=0) = \tilde{T}_0$

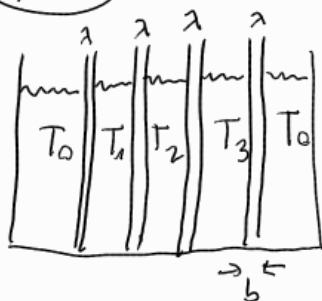
$$\tilde{T}(x \rightarrow \infty) = 0$$

$k > 0$ isto kot pri 2)

$k < 0$ (vzameneno $\rightarrow$) $\checkmark$

Za splošne robne pogoje: $\tilde{T}(x) = Ae^{k_1 x} + Be^{k_2 x}$

7.16 1. izpit 2019



$T(t) = ?$